

Work Stream 4: Cavitation as the First Constitutive Limit in Liquids

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1 Introduction

u/Little-Name9809 asked whether the OpenAI construction suggests a way to drive a liquid to extreme states using *only mechanical force*. Working through the question gives a sharper result than the one originally anticipated: for a liquid at ordinary pressure, the first physical assumption of the incompressible model to fail is not incompressibility, not the continuum hypothesis, and not viscous heating, but the liquid’s inability to sustain tension. Cavitation comes first, and by a wide margin.

2 Hydrodynamic Cavitation

In the OpenAI proof, the collapsing core has velocity $|u(\tau)| \sim \tau^{-1/2-h}$ with $h = 1/200$, and radius $\ell_r \sim \tau^{1/2}$. Because the core keeps a radial Reynolds number of order one, these scales are diffusive in any units, $\ell_r \simeq \sqrt{\nu t}$ and $u \simeq \sqrt{\nu/t}$, essentially independent of the initial vortex size. The static pressure at the centre of a vortex lies below the ambient pressure by roughly the dynamic pressure, $\Delta p \sim \rho u^2$ (exactly $\rho u_{\max}^2$ for a Rankine vortex). The local cavitation number

$$\sigma(\tau) = \frac{p_\infty - p_v}{\frac{1}{2}\rho|u(\tau)|^2} \quad (1)$$

therefore falls to order one when $\rho u^2 \sim p_\infty - p_v$.

Table 1: Where the liquid fails first: water at 300 K, $\nu = 10^{-6} \text{ m}^2/\text{s}$, $c = 1500 \text{ m/s}$. Times are physical times before blow-up, $t \simeq \nu/u^2$; the τ^{-h} corrections change them by a factor $\lesssim 1.3$.

Limit	u (m/s)	Ma	t before blow-up	ℓ_r
Cavitation, $p_\infty - p_v \approx 0.1 \text{ MPa}$ (water with nuclei)	14	0.009	5 ns	70 nm
Tensile strength $\approx 30 \text{ MPa}$ (clean samples)	245	0.16	17 ps	4 nm
Tensile strength $\approx 140 \text{ MPa}$ (inclusions, near spinodal)	530	0.35	3.6 ps	2 nm
Compressibility, $\text{Ma} = 0.3$	450	0.3	5–7 ps	3 nm
Viscous heating to boiling, $\Delta T = u^2/c_p = 73 \text{ K}$	550	0.37	3–4.5 ps	2 nm

For ordinary water containing nucleation sites, cavitation sets in at $u \approx 14 \text{ m/s}$, about 5 ns before the mathematical blow-up, when the core is still $\approx 70 \text{ nm}$ across: three orders of magnitude in time before the Mach limit and while the continuum description is unquestionably valid. Even the metastable tensile limits of very clean water are reached before, or together with, $\text{Ma} = 0.3$. The Bernoulli pressure drop, not thermal shock, is the mechanism that removes a real liquid from the incompressible model.

Caveats. The core at these times is tens of nanometres wide and lives for nanoseconds, so nucleation kinetics (the number of nuclei in a 70 nm region, the time to grow a critical bubble) matter and can delay cavitation toward the tensile-strength rows of the table. The comparison with the thermal estimate of Work Stream 2 is consistent: viscous heating gives $\Delta T = u^2/c_p$, which is only 48 K at $\text{Ma} = 0.3$; there is no “plasma by pure force”—temperatures of order 10^4 K would require $\text{Ma} \gtrsim 4$, far outside the model’s validity.

Speculative remark. One might ask whether the violent collapse of the resulting cavity could concentrate energy in the manner of sonoluminescence. We record this as an unsupported analogy only: nothing in the present analysis establishes it, and the construction provides no blueprint for it.

3 Acknowledgments

Thank you to `u/Little-Name9809` for connecting the pure-mathematics PDE construction to the physics of phase transitions; the question led to a cleaner ordering of the validity limits than the thermal argument alone.